

CityTech TTP Summer Bootcamp

Introductions, Environment Setup, and GitHub

2026-06-01

Agenda

1. git Setup and Repo Review
2. Introductions
3. Range Finding
4. Review
5. Environment Setup

git Setup and Repo Review

- Get onto WiFi
- Create / sign into GitHub
- Bootcamp repo is here: <https://github.com/jonathan-chin/citytech-ttpr-2026-summer>
- Follow along!

Introductions

- What is TTP?
- Who am I?
- Who are the TAs?
- Who is the Program Director?

This or That? — Pick a side and defend it

- 🍏 Mac or 🖥️ Windows?
- 🧊 Iced coffee or ☕ Hot coffee?
- 🎤 Karaoke or 🙋 Hard pass?
- 🏙️ Manhattan or 🌉 Brooklyn?
- 🌶️ Spicy or 😊 Mild?
- 📖 Read the book or 🎧 Audiobook?

This or That? — Favorites and hot takes

-  Marvel or  DC?
-  Burrito or  Tacos?
-  Soccer or  Basketball?
-  Horse-sized duck or  100 duck-sized horses?
-  Nintendo or  PlayStation?

Range Finding

Let's see what you already know — shout it out

Range Finding

- What is an **SSH key**?
- What are **Copilot, Claude, and Gemini**?
- What is **git**?

Range Finding

- What is the difference between an **app**, a **web app**, and a **hybrid app**?
- What is **responsive design**?
- What is **Web Inspector / DevTools**?

Range Finding

- What is an **API**, in general?
- What is an **OAuth token**?
- What is **input sanitation**?

Range Finding

- What is a **database**?
- What is **SQL**?
- What is a **server**?

Range Finding

- What is an **SPA** (single-page application)?
- What is a **component**, in React for example?
- What is **CORS**?

Range Finding

- What is **web scraping**?
- What is **exploratory data analysis**?
- What is **PII**?

Range Finding

- What are **3 kinds of data visualization charts?**

Range Finding

- What is a **password hash**?
- What is **authentication**?
- What is **authorization**?

Review

- What do you think we'll be learning this summer?
- What are you most excited for?

Environment Setup

Let's get your machines ready — follow along

macOS — start with Homebrew · **Windows** — start by enabling WSL2

1 · Homebrew (macOS package manager)

The easiest way to install most of today's tools on a Mac.

- Paste into Terminal:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

- **Windows** — no Homebrew; set up WSL2 next (slide 2), then use `apt` inside WSL.

Guide: <https://brew.sh>

2 · WSL2 (Windows)

WSL (Windows Subsystem for Linux) runs a real Linux environment — Ubuntu — right on Windows, giving you the same Unix shell, tools, and `apt` package manager that macOS and Linux users have.

- **Windows** — in an **Admin PowerShell**, then reboot:

```
wsl --install
```

From here on, run your command-line installs inside the WSL (Ubuntu) terminal — follow the macOS/Linux steps, not PowerShell or Windows `.exe` installers.

Guide: <https://learn.microsoft.com/windows/wsl/install>

3 · Node, nvm and Yarn

nvm — copy the install `curl` command from the nvm README and run it:

```
# use the current version shown in the nvm README
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
```

Node — `nvm install stable` · **Yarn** — `corepack enable` (per yarnpkg.com)

Note: Yarn 1.x ("Classic") is the legacy version in many tutorials; 4.x ("Berry") is current — Corepack gives you the modern one.

Windows — run all of this inside **WSL**.

nvm: <https://github.com/nvm-sh/nvm> · Yarn: <https://yarnpkg.com/getting-started/install>

4 • Git

- **macOS** — `brew install git`
- **Windows** — inside **WSL**, follow the Linux install: `sudo apt install git`

Verify with `git --version`, then set your identity:

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"
```

Linux / install guide: <https://git-scm.com/download/linux>

5 · VS Code

Our editor for the whole bootcamp.

- **macOS** — `brew install --cask visual-studio-code` or download the `.dmg`
- **Windows** — run the installer, then add the **WSL** extension to work inside Linux

Recommended extensions: **WSL**, **GitHub Copilot**, **ESLint**, **Prettier**, **Python**.

Download: <https://code.visualstudio.com/download>

6.1 · SSH Key — Do you already have one?

Lets you push to GitHub without passwords. **Check before making a new one.**

- List existing keys:

```
ls -al ~/.ssh
```

- Look for a **public** key file like `id_ed25519.pub` or `id_rsa.pub`.
- If you see one, you can reuse it — skip to adding it to GitHub on the next slide.

Checking for keys: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/checking-for-existing-ssh-keys>

6.2 · SSH Key — Create one and add to GitHub

- Generate a key (press Enter to accept the defaults):

```
ssh-keygen -t ed25519 -C "you@example.com"
```

- Copy the **public** key:

```
# macOS  
pbcopy < ~/.ssh/id_ed25519.pub  
# Linux / WSL – print it, then copy the output  
cat ~/.ssh/id_ed25519.pub
```

- Paste it into **GitHub** → **Settings** → **SSH and GPG keys** → **New SSH key**.

Generating and adding a key: <https://docs.github.com/en/authentication/connecting-to-github-with-ssh>

7 · Insomnia (API client)

For testing APIs.

- **macOS** — `brew install --cask insomnia` or download
- **Windows** — run the installer

Download: <https://insomnia.rest/download>

8 · PostgreSQL

Our database.

- **macOS** — `brew install postgresql@16` (or **Postgres.app**)
- **Windows** — run the **EDB installer** (bundles the server + tools)

Download: <https://www.postgresql.org/download/>

9 · pgAdmin

A GUI for browsing and querying PostgreSQL.

- **macOS** — `brew install --cask pgadmin4` or download
- **Windows** — run the installer (or it's bundled with the EDB installer above)

Download: <https://www.pgadmin.org/download/>

10 · Python 3

For data science.

- **macOS** — `brew install python`
- **Windows** — inside **WSL**, follow the Linux install: `sudo apt install python3 python3-pip`

Verify with `python3 --version`.

Download / install guide: <https://www.python.org/downloads/>

11 · Jupyter

Notebooks for data exploration and visualization.

Once Python is installed (Windows: inside **WSL**):

```
pip3 install jupyterlab  
jupyter lab
```

Guide: <https://jupyter.org/install>

12 · Microsoft 365 and Copilot

- Log in to **Microsoft 365** with your **CUNY credentials**
- Make sure you can access **Copilot** — tell it "Hello World"

CUNY Microsoft 365: <https://www.cuny.edu/about/administration/offices/cis/technology-services/microsoft-office-365-for-education/>

Sign in: <https://www.microsoft365.com>

Copilot: <https://copilot.microsoft.com>